GRAND CANYON NATIONAL PARK
WORLD HERITAGE SITE
REHABILITATE NORTH RIM INFRASTRUCTURE
VOLUME 12 POWER SYSTEM – COTTONWOOD CAMPGROUND

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 3	PIPELINES - NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES - ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 6	NORTH RIM WATER TREATMENT PLANT
VOLUME 7	NOT USED
VOLUME 8	NORTH RIM WASTEWATER COLLECTION SYSTEM
VOLUME 9	POWER DISTRIBUTION - NORTH RIM
VOLUME 10	POWER DISTRIBUTION - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	120G001	GENERAL	SHEET INDEX
2	121X101	PHOTOVOLTAIC SYSTEM	DEMOLITION PLAN
3	121C101	PHOTOVOLTAIC SYSTEM	SITE PLAN
4	121S001	PHOTOVOLTAIC SYSTEM	SOLAR SHADE CANOPY GENERAL NOTES
5	121S101	PHOTOVOLTAIC SYSTEM	SOLAR SHADE CANOPY FOUNDATION AND ROOF PLANS
6	121S501	PHOTOVOLTAIC SYSTEM	SOLAR SHADE CANOPY SECTION AND DETAILS
7	121A101	PHOTOVOLTAIC SYSTEM	SOLAR SHADE CANOPY ARCHITECTURAL PLAN AND SECTION
8	121E501	PHOTOVOLTAIC SYSTEM	GROUNDING AND TRENCH DETAILS
9	121E502	PHOTOVOLTAIC SYSTEM	SOLAR / BATTERY DETAILS
10	121E503	PHOTOVOLTAIC SYSTEM	PV CONNECTION DETAILS
11	121E601	PHOTOVOLTAIC SYSTEM	DC / AC ONE-LINE DIAGRAM

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO. 75014
TOU VANG
Date Signed
STATE OF ARIZONA

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 41909
RYAN RHOADES
Date Signed
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. KUCHLE CADD N. LYNCH TECH. REVIEW: M. GOUGH DATE: AUG 2026	SUB SHEET NO. 120 G001	TITLE OF SHEET GENERAL SHEET INDEX NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 1 OF 11

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KEY NOTES: (X)

1. CONTRACTOR TO CLEAR AND GRUB AREA FOR SHADE STRUCTURE INSTALLATION. VERIFY FINAL SHADE STRUCTURE LOCATION WITH CONTRACTING OFFICER REPRESENTATIVE BEFORE CLEARING AREA.
2. EXIST SOLAR PHOTOVOLTAIC ARRAY TO BE REMOVED. REMOVE BURIED CABLE AND CONDUIT BACK TO SOURCE. COORDINATE WITH CONTRACTING OFFICER FOR SALVAGE OF PV PANELS AND INVERTERS.
3. PROTECT FIRE HOSE BOX IN PLACE.
4. CONTRACTOR TO CLEAR AND GRUB AREA FOR BATTERY ENCLOSURE AND INVERTER. VERIFY FINAL LOCATION WITH CONTRACTING OFFICER REPRESENTATIVE BEFORE CLEARING AREA.
5. AVAILABLE CONSTRUCTION LIMITS ARE SHOWN IN THE LIMITS OF CONSTRUCTION TABLE.

LIMITS OF CONSTRUCTION		
POINT NO.	NORTHING	EASTING
1201	13134259.71	1333333.83
1202	13134222.12	1333276.76
1203	13134209.55	1333232.83
1204	13134185.02	1333190.83
1205	13134183.75	1333166.57
1206	13134231.04	1333146.52
1207	13134238.90	1333163.64
1208	13134282.73	1333224.38
1209	13134312.86	1333298.83

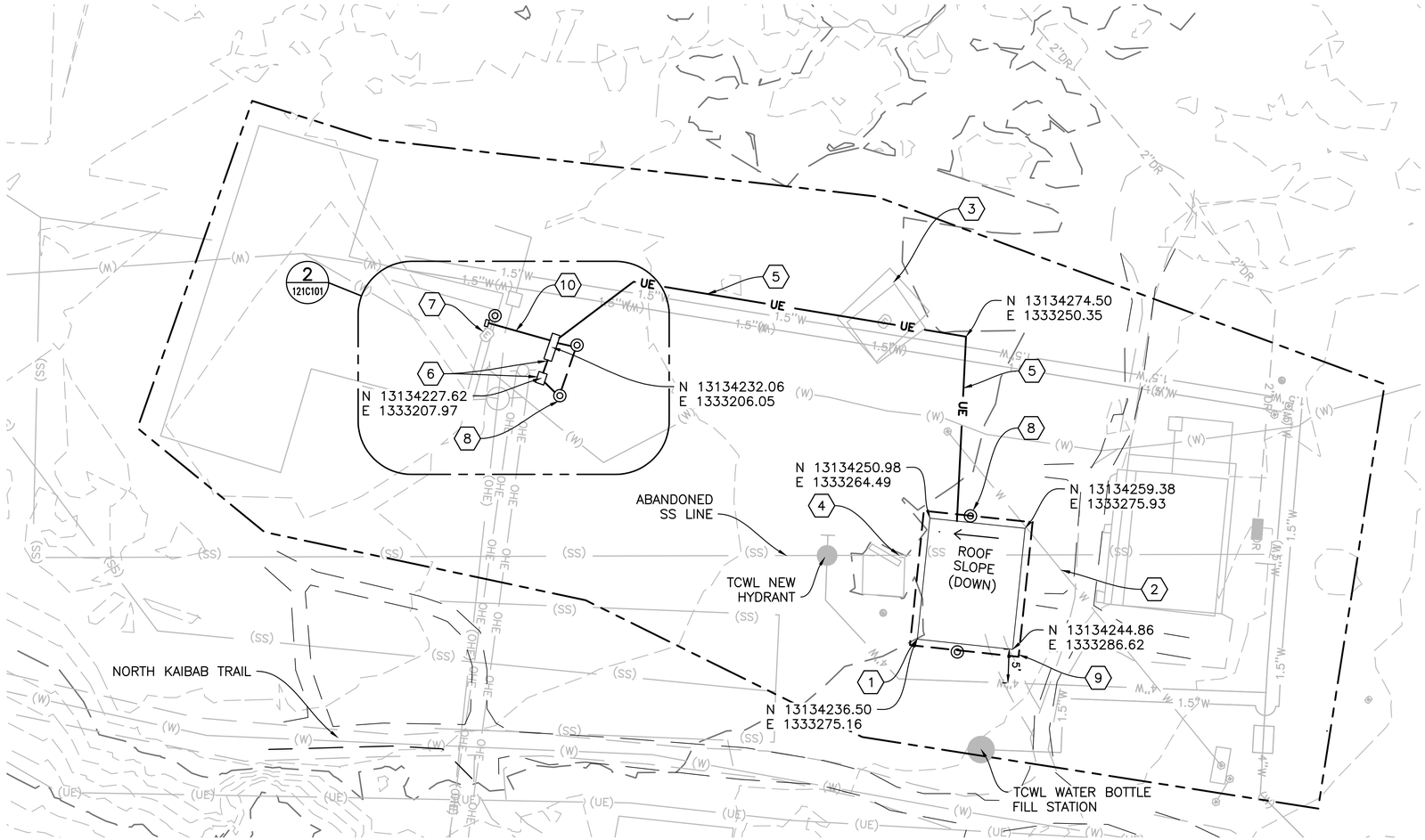
10 0 10 20
SCALE OF FEET
1" = 10'

REGISTERED PROFESSIONAL ENGINEER ELECTRONICS
CERTIFICATE NO.
75014
TOU
VANG
STATE OF COLORADO

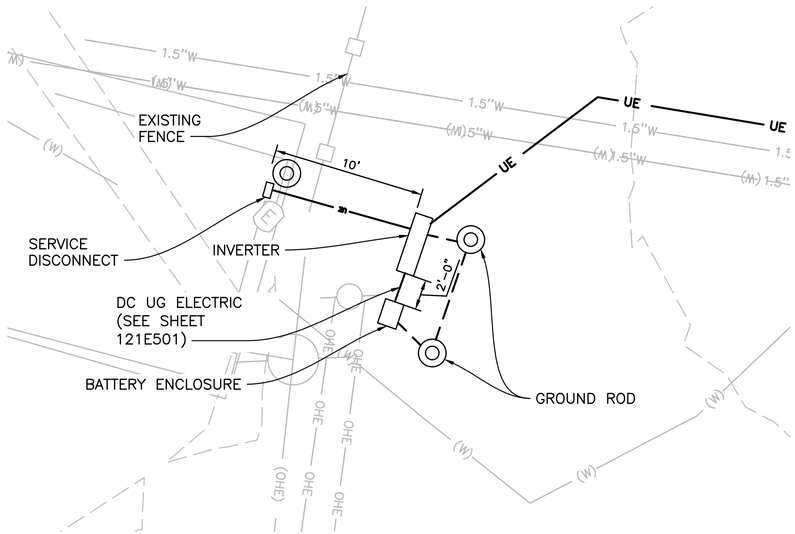
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. KUCHLE CADD D. VANG TECH. REVIEW: A. KANER DATE: MAY 2026	SUB SHEET NO. 121 X101	TITLE OF SHEET PHOTOVOLTAIC SYSTEM DEMOLITION PLAN NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113
182752
PMIS/PKG NO. 318719
SHEET 2 OF 11

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1 121C101 RANGER STATION
SCALE (A)



2 121C101 ENLARGED VIEW – BATTERY STORAGE UNIT AT RANGER STATION
SCALE (B)

KEY NOTES: (X)

1. SOLAR SHADE STRUCTURE.
2. FIELD VERIFY AND REROUTE EXIST WATERLINE AS NEEDED FOR SHADE STRUCTURE.
3. SEE DEMO SHEET 121X101.
4. CONTRACTOR TO FIELD LOCATE EXIST UTILITIES IN SHADE STRUCTURE AREA. EXPOSE AND CAP UTILITIES AS NEEDED AND ABANDON IN PLACE.
5. CONTRACTOR TO FIELD LOCATE EXISTING PV CONDUIT. SHADE STRUCTURE DC CONDUIT SHALL FOLLOW EXIST PV CONDUIT TRENCH WHERE POSSIBLE. SEE SHEET 121E501 FOR UE CABLE AND CONDUIT TRENCH DETAILS.
6. BATTERY AND INVERTER LOCATION. INVERTER AND CHARGE CONTROLLER SHALL BE INSTALLED ON MODULAR EQUIPMENT RACK. SEE SHEET 121E502 FOR MODULAR EQUIPMENT RACK AND BATTERY ENCLOSURE DETAILS. SEE 121S501 FOR PAD DETAILS.
7. EXIST SERVICE DISCONNECT. SEE SHEET 121E502 FOR SERVICE DISCONNECT INSTALLATION DETAILS.
8. GROUND ROD. SEE SHEET 121E501 FOR DETAILS.
9. GROUNDING RING. SEE SHEET 121E501 FOR DETAILS.
10. AC UE TRENCH FROM INVERTER TO SERVICE DISCONNECT. SEE SHEET 121E501 FOR TRENCH DETAILS.

SCALE (B) 5 0 5 10 1" = 5'
SCALE OF FEET

SCALE (A) 10 0 10 20 1" = 10'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. KUCHLE DRAWN: D. VANG TECH. REVIEW: A. KANER DATE: MAY 2026	SUB SHEET NO. 121 C101	TITLE OF SHEET PHOTOVOLTAIC SYSTEM SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 3 OF 11

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GENERAL

G1. SCOPE
THE NOTES ON THIS SHEET AND THE STANDARD STRUCTURAL DETAILS ARE GENERAL AND APPLY TO THE ENTIRE PROJECT WHETHER SPECIFICALLY CALLED OUT OR NOT, EXCEPT WHERE THERE ARE SPECIFIC INDICATIONS TO THE CONTRARY ON STRUCTURAL SHEETS. IF THERE ARE QUESTIONS, THEY SHALL BE SUBMITTED TO THE CONTRACTING OFFCIER AND ANSWERED IN WRITING PRIOR TO CONSTRUCTION.

G2. APPLICABLE SPECIFICATIONS AND CODES
A. INTERNATIONAL BUILDING CODE, IBC 2024, WITH APPLICABLE EDITIONS OF THE CODE REFERENCED STANDARDS.
B. AUTHORITY HAVING JURISTICTION: NATIONAL PARK SERVICE (WIND, SNOW, AND SEISMIC DESIGN CRITERIA IN ACCORDANCE WITH THE MORE CONSERVATIVE OF COCONINI COUNTY, AZ AND NATIONAL PARK SERVICE).
C. BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE, ACI 318-19.
D. MINIMUM DESIGN LOADS AND ASSOCIATED CRITERIA FOR BUILDINGS AND OTHER STRUCTURES, ASCE 7-22.
E. AMERICAN INSTITUTE OF STEEL CONSTRUCTION STEEL CONSTRUCTION MANUAL, 15TH ED.
F. AMERICAN WELDING SOCIETY, AWS D1.1.
G. ASTM MATERIAL STANDARDS AS NOTED.

G3. DESIGN CRITERIA
1. APPLIES TO ALL STRUCTURES
A. DEAD LOAD:
1. ACTUAL TRIBUTARY STRUCTURE WEIGHT
2. SUPERIMPOSED DEAD LOAD: 3 PSF (SOLAR PANELS)
B. LIVE LOAD:
1. ROOF: 20 PSF (NOT REDUCIBLE)
C. WIND:
1. ULTIMATE WIND SPEED (V_{ULT}): 101 MPH
2. TORNADO SPEED (V_T): NOT APPLICABLE PER ASCE 7 FIGURES 32.1-1 & 1-2
3. NOMINAL WIND SPEED (V_{ASD}): 79 MPH
4. RISK CATEGORY: II
5. EFFECTIVE PLAN AREA (A_E): NOT APPLICABLE
6. EXPOSURE CATEGORY: C
7. ENCLOSURE CLASSIFICATION: OPEN

D. SEISMIC:
1. ABOVE GRADE, CANOPY STRUCTURE:
a. RISK CATEGORY, II
b. IMPORTANCE FACTOR: 1.00
c. SPECTRAL RESPONSE ACCELERATION, SS = 0.42 g
d. SPECTRAL RESPONSE ACCELERATION, S1 = 0.11 g
e. SITE CLASS: D
f. SEISMIC DESIGN CATEGORY: C
g. SPECTRAL RESPONSE ACCELERATION, SDS = 0.38 g
h. SPECTRAL RESPONSE ACCELERATION, SD1 = 0.2 g
i. BASIC SEISMIC FORCE RESISTING SYSTEM: STEEL SYSTEMS NOT SPECIFICALLY DETAILED FOR SEISMIC RESISTANCE,
j. ANALYSIS PROCEDURE: EQUIVALENT LATERAL FORCE
k. DESIGN BASE SHEAR: 0.10 KIPS
l. SEISMIC RESPONSE COEFFICIENT, Cs = 0.07
m. RESPONSE MODIFICATION FACTOR, R = 3.0

E. SNOW LOAD:
1. GROUND SNOW LOAD = 62 PSF
2. SLOPED ROOF SNOW LOAD = 44 PSF
3. EXPOSURE FACTOR = 0.90
4. IMPORTANCE FACTOR = 1.00
5. THERMAL FACTOR = 1.20
6. FROST DEPTH: = 24 IN
7. WINTER WIND PARAMETER (W_2) = 0.35

G4. THE FOLLOWING NON-CONTRACTUAL GEOTECHNICAL REPORT WAS DEVELOPED FOR THIS PROJECT AND IS THE BASIS OF THIS STRUCTURAL DESIGN:
GEOTECHNICAL FIRM NAME: LITHOS ENGINEERING
ADDRESS: 2750 S. WADSWORTH BLVD, SUITE D-200 DENVER, CO 80227
REPORT NUMBER: 21186
REPORT DATE: JANUARY 30, 2024
NET ALLOWABLE BEARING PRESSURE: 2000 PSF

G5. SAFETY
SAFETY AND STRUCTURE STABILITY DURING CONSTRUCTION ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR. STRUCTURES HAVE BEEN DESIGNED TO RESIST THE DESIGN LIVE LOADS ONLY AS A COMPLETED STRUCTURE.

G6. OPENINGS
OPENINGS FOR PIPES, DUCTS, CONDUITS, ETC. ARE NOT ALL SHOWN ON THE STRUCTURAL DRAWINGS. COORDINATE AND PROVIDE OPENINGS AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

G7. SPECIAL INSPECTIONS
SPECIAL INSPECTIONS ARE REQUIRED IN ACCORDANCE WITH CHAPTER 1 AND CHAPTER 17 OF THE IBC. THE CONTRACTOR IS RESPONSIBLE FOR SPECIAL INSPECTIONS (STSI) BY QUALIFIED PERSONNEL. THE CONTRACTOR SHALL PROVIDE FOR FULL ACCESS TO THE WORK BY STSI AGENTS AND SHALL PROVIDE FOR THESE INSPECTIONS IN THEIR CONSTRUCTION SCHEDULE IN ACCORDANCE WITH THE SPECIFICATIONS. THE SPECIAL INSPECTION PROGRAM AND STATEMENT OF SPECIAL INSPECTIONS SHALL BE IN ACCORDANCE WITH 01 45 33.

G8. STANDARD DETAILS
THE STANDARD DETAILS DEPICT TYPICAL DETAILING TO BE USED ON THIS PROJECT. IF CONDITIONS ARE NOT EXPLICITLY SHOWN ON THE DRAWINGS THEY SHALL BE MADE SIMILAR TO THE STANDARD DETAILS. OBTAIN APPROVAL OF THE CONTRACTING OFFICER IN WRITING FOR SIMILAR CONDITIONS PRIOR TO CONSTRUCTION.

G9. FIELD VERIFY ALL DIMENSIONS AND ELEVATIONS OF EXISTING CONSTRUCTION AS REQUIRED TO COORDINATE NEW CONSTRUCTION. SUBMIT REQUIRED CHANGES FOR APPROVAL.

G10. SUBMIT FOR REVIEW ALL EQUIPMENT SIZES, OPERATING WEIGHTS, VIBRATION FORCES, SUPPORT LOCATIONS, ALONG WITH ANY FLOOR OPENINGS, NOTCHES, AND RECESSES REQUIRED BY SUCH EQUIPMENT. CONCRETE SUPPORT PADS AND/OR FRAMING REQUIRED TO SUPPORT SAID EQUIPMENT SHALL NOT BE FABRICATED AND PLACED UNTIL THE CONCRETE SUPPORT PADS AND/OR FRAMING IS APPROVED TO SUPPORT THE EQUIPMENT.

CONCRETE

C1. DESIGN STRENGTHS:
F'c = 4,500 PSI
REBAR Fy = 60,000 PSI
PRECAST F'c = 5,000 PSI

C2. CONCRETE COVER
UNLESS OTHERWISE NOTED, PROVIDE CONCRETE COVER FOR REINFORCING AS FOLLOWS:
CONCRETE DEPOSITED AGAINST EARTH: 3"
ALL OTHER: 2"
WATERSTOPS: 3"
SEE DRAWINGS FOR EXCEPTIONS

C3. SEE SPECIFICATIONS FOR REINFORCING PLACEMENT REQUIREMENTS.

C4. REFER TO OTHER DISCIPLINE DRAWINGS PRIOR TO CONSTRUCTION FOR EMBEDDED ITEMS AND PENETRATIONS NOT SHOWN ON STRUCTURAL DRAWINGS. AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

C5. PROVIDE 3/4" CHAMFERS AT ALL EXPOSED EDGES. NOT ALL CHAMFERS MAY BE SHOWN ON DRAWINGS.

C6. FIELD ADJUST REINFORCING AT OPENINGS AND EMBEDDED ITEMS AS INDICATED.

C7. ANCHOR BOLTS NOT SPECIFIED BY CONTRACTING OFFICER SHALL BE DESIGNED AND CERTIFIED BY A REGISTERED PROFESSIONAL ENGINEER LICENSED IN THE STATE OF ARIZONA, RETAINED BY THE CONTRACTOR, IN ACCORDANCE WITH APPLICABLE PROJECT AND CODE REQUIREMENTS. SUBMIT AS A SHOP DRAWING FOR REVIEW AND APPROVAL BY THE CONTRACTING OFFICER. COORDINATE LOCATION, SIZE AND EMBEDMENT PRIOR TO CASTING CONCRETE.

C8. CONTINUOUS WATERSTOP SHALL BE INSTALLED IN JOINTS SUBJECT TO STATIC WATER PRESSURE.

C9. ABSOLUTELY NO WELDING OF REINFORCING BARS OR TORCHING TO BEND REINFORCING BARS SHALL BE ALLOWED WITHOUT SPECIFIC APPROVAL FROM THE CONTRACTING OFFICER.

C10. CONTRACTOR SHALL SUBMIT A CONCRETE PLACEMENT PLAN PER SPECIFICATION 03 31 31 IDENTIFYING JOINT TYPES, JOINT LOCATIONS AND CONCRETE PLACEMENT SEQUENCE.

C11. ALL CAST IN PLACE AND POST-INSTALLED ANCHORS INDICATED IN THE STRUCTURAL DOCUMENTS SHALL COMPLY WITH CHAPTER 17 OF ACI 318 AND CHAPTER 19 OF THE IBC. ALL EXPANSION AND ADHESIVE ANCHORS SHALL HAVE THE ICC REPORT SHOWING EQUIVALENT LOAD CAPACITY. SUBMIT AND INSTALL PER THE ICC EVALUATION REPORT.

STEEL

S1. DESIGN STRENGTHS:
WIDE FLANGE AND TEES: Fy=50 KSI
PIPES: Fy=35 KSI
STAINLESS STEEL: Fy=33 KSI
HSS SECTIONS: Fy=50 KSI
ALL OTHER PLATES AND SHAPES: Fy=36 KSI
ANCHOR RODS: Fy=36 KSI

S2. DIMENSIONS:
TO CENTERLINES OF COLUMNS AND BEAMS, TOP SURFACES OF BEAMS AND TUBES AND BACKS OF CHANNELS AND ANGLES UNO.

S3. ELEVATIONS:
TOP OF STEEL REFERS TO TOP SURFACE OF MEMBER OR FLANGE UNO.

S4. WHEN FILLET WELD SIZE IS NOT INDICATED, PROVIDE MAXIMUM WELD SIZE BASED ON MATERIAL THICKNESS IN ACCORDANCE WITH AISC SPECIFICATIONS.

S5. ALL BOLTED STRUCTURAL CONNECTIONS ARE BEARING TYPE CONNECTIONS UNLESS OTHERWISE SPECIFIED TO BE SLIP-CRITICAL. PROVIDE LOAD INDICATING WASHERS AT SLIP-CRITICAL CONNECTIONS.

S6. CONFORM TO AISC 360, STEEL CONSTRUCTION MANUAL AND AISC 341, SEISMIC DESIGN MANUAL.

S7. THE SEISMIC LOAD RESISTING SYSTEM (SLRS) IS DENOTED ON THE FRAMING PLANS AND FRAME ELEVATIONS. THE SLRS DESIGNATION INCLUDES THE MEMBER AND CONNECTIONS AT EACH END. FRAMES ARE DENOTED AS FOLLOWS:

A. MOMENT FRAME: (SLRS) ◄——►

FOR SLRS SYSTEMS OR PARTS OF SYSTEMS THAT ARE NOT INCLUDED IN FRAMES, (SLRS) IS PLACED NEXT TO THE BEAM SIZE ON THE FRAMING PLAN (eg: ON A COLLECTOR ELEMENT (DRAG STRUT):

S8. ALL STEEL BEAMS SHALL RECEIVE STANDARD CAMBER PER THE SPECIFICATIONS UNLESS NOTED OTHERWISE ON THE PLANS. BEAMS REQUIRING SPECIAL CAMBER ARE DENOTED ON THE BEAMS SHOWN ON THE FRAMING PLANS. EXAMPLE: (+1/2") INDICATES ½".

S9. PROVIDE 1/4" CAP PL AT ALL OPEN ENDS OF HSS FRAMING MEMBERS AND SEAL WELD.

EARTHWORK

E1. ALL EARTHWORK AND SITE PREPARATION SHALL BE PERFORMED IN STRICT ACCORDANCE WITH THE SPECIFICATIONS. ALL FOUNDATION EXCAVATIONS SHALL BE OBSERVED AND APPROVED BY THE GEOTECHNICAL ENGINEER.

E2. ANY EXISTING FILLS OR UNSUITABLE SOILS AS DETERMINED BY THE GEOTECHNICAL ENGINEER SHALL BE EXCAVATED AND REPLACED WITH PROPERLY COMPACTED SELECT FILL PER SPECIFICATIONS.

E3. CONTRACTOR SHALL EXERCISE EXTREME CARE WHEN EXCAVATING OR GRADING ADJACENT TO EXISTING STRUCTURES SO AS TO NOT DAMAGE OR UNDERMINE FOUNDATIONS, WALLS, SLABS, OR UTILITIES, ETC.

E4. DO NOT EXCAVATE BELOW THE BEARING ELEVATION OF ANY COMPLETED FOOTING NOR ANY CLOSER TO THE FOOTING THAN A SLOPE OF TWO HORIZONTAL (MEASURED FROM EDGE OF FOOTING TO NEAREST POINT IN EXCAVATION) TO ONE VERTICAL.

E5. ALL FOUNDATIONS AND SLABS ON GRADE SHALL BE PLACED ON A MINIMUM OF 12" OF SELECT FILL PER SPECIFICATIONS.

DELEGATED SUBMITALS

D1. DEFINITION:
DELEGATED DESIGN SUBMITTALS ARE ITEMS FOR WHICH THE FINAL DESIGN OF A SPECIFIC ELEMENT OF THE PERMANENT CONSTRUCTION WORK EXPRESSLY DELEGATED TO A PROPERLY LICENSED AND REGISTERED DESIGN PROFESSIONAL RETAINED OR EMPLOYED BY THE CONTRACTOR, SUBCONTRACTOR, SUPPLIER OF ANY TIER. "DELEGATED DESIGN" APPLIES ONLY TO PERMANENT WORK THAT WILL BE PART OF THE COMPLETED PROJECT AS A FUNCTIONING WHOLE.

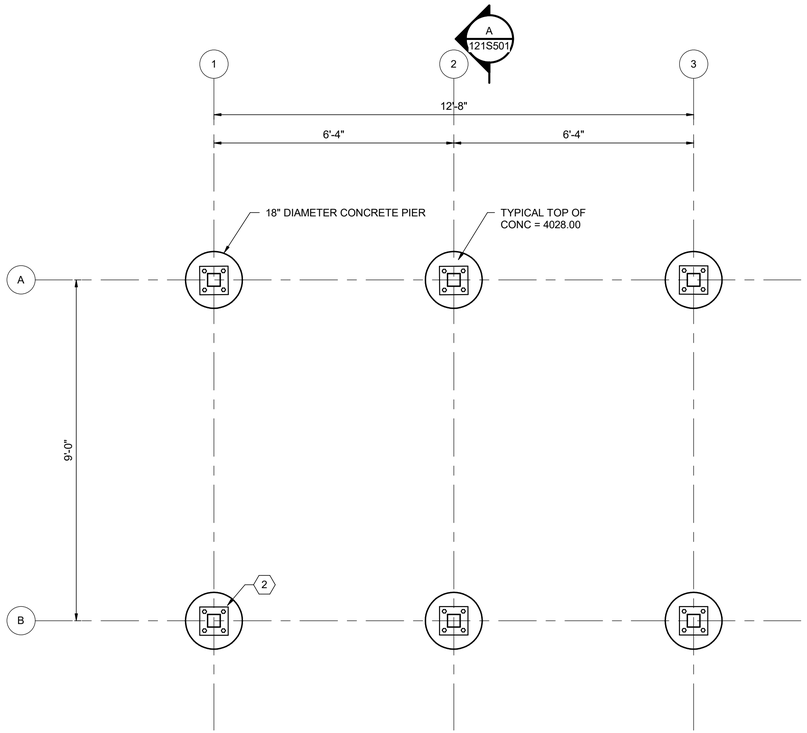
D2. DELEGATED DESIGN LIST:
THE FOLLOWING IS A LIST OF DELEGATED DESIGN SUBMITTALS THAT ARE EXPECTED TO CONTAIN STRUCTURAL CALCULATIONS OF SAFETY-RELATED SYSTEM INFORMATION FOR REVIEW. THESE SUBMITTALS APPLY TO DELEGATED DESIGNED SYSTEMS PRIOR TO INSTALLATION OF THE INDICATED STRUCTURAL ELEMENT, EQUIPMENT, DISTRIBUTION SYSTEM, COMPONENT, OR ITS ANCHORAGE. THE CONTRACTOR SHALL SUBMIT THE REQUIRED ENGINEER CERTIFICATION, SUPPORTING DATA, AND DRAWINGS FOR REVIEW AND ACCEPTANCE BY THE CONTRACTING OFFICER.

D3. DELEGATED DESIGN SUBMITTAL INCLUDE, BUT ARE NOT LIMITED TO, THE FOLLOWING:
1. SOLAR PANELS
2. SOLAR PANEL SUPPORTS

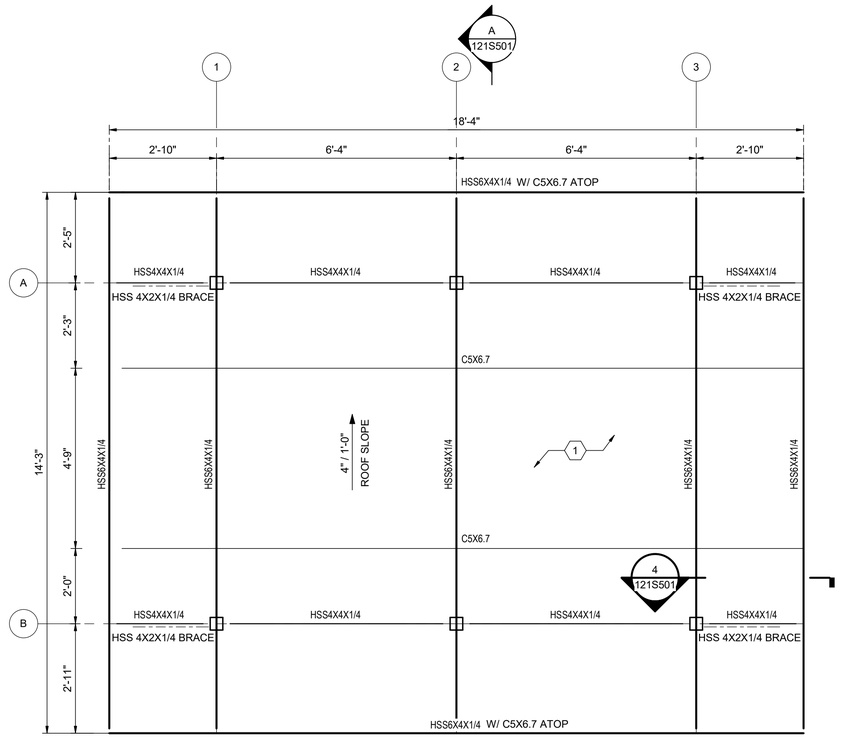
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REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 85509 — JOSEPH CORONADO — ARIZONA, U.S.A. — Date signed 4/17/2026

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. PEREZ DRAWN: J. PEREZ TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 121 S001	TITLE OF SHEET PHOTOVOLTAIC SYSTEM SOLAR SHADE CANOPY GENERAL NOTES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 4 OF 11



1 121S101 FOUNDATION PLAN
SCALE A



2 121S101 ROOF PLAN
SCALE A

GENERAL NOTES

A. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING. EQUIPMENT, ANCHORS, CONDUITS, HOUSEKEEPING PADS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTURAL ELEMENTS THAT MAY NOT BE SHOWN.

B. ALL STEEL SHALL BE SHOP PRIMED AND PAINTED IN ACCORDANCE WITH SPEC. 09 96 00.

KEYNOTES

1 CORRUGATED CORETEN ROOF PER ARCHITECTURE. FASTENING PER MFR.

2 TYPICAL COLUMN AND BASE PLATE:HSS 4x4x1/4 W/ BASE PL3/4x9x9 W/ (4) 1/2" DIA x 12" EMBEDMENT ANCHOR RODS. RODS TO BE F1554, GR 36, GALVANIZED.

SCALE A 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 85509
JUAN CORONADO
Date Signed electronically
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. PEREZ DRAWN: J. PEREZ TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 121 S101	TITLE OF SHEET PHOTOVOLTAIC SYSTEM SOLAR SHADE CANOPY FOUNDATION AND ROOF PLANS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 5 OF 11

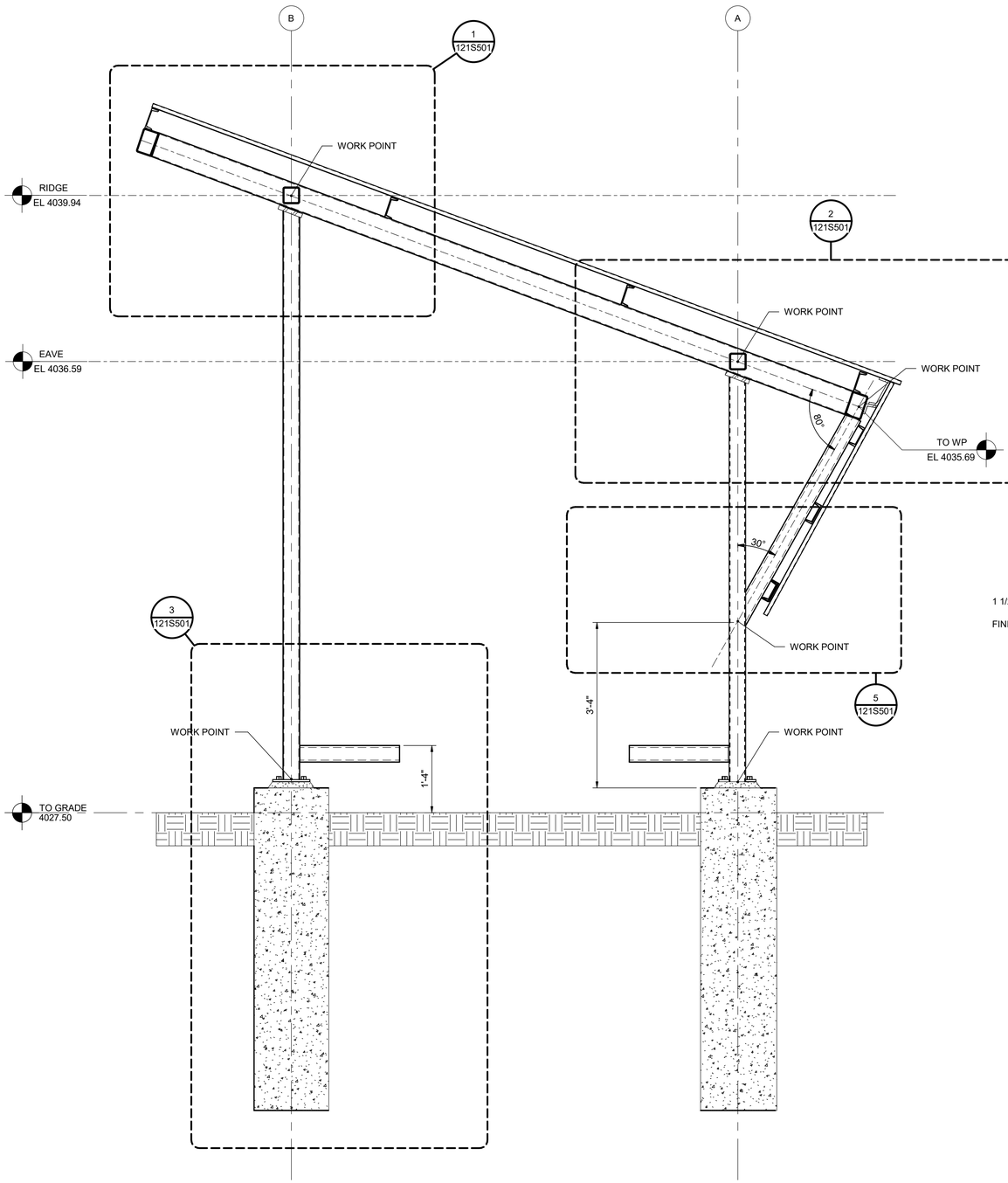
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GENERAL NOTES

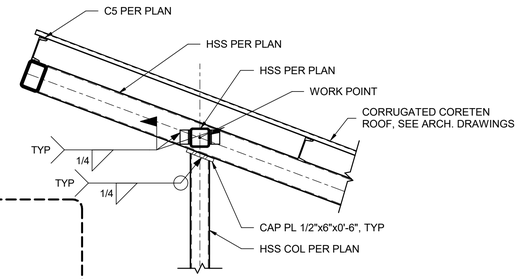
A. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING. EQUIPMENT, ANCHORS. CONDUITS, HOUSEKEEPING PADS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTURAL ELEMENTS THAT MAY NOT BE SHOWN.

B. ATTACHMENT OF SOLAR ARRAY TO STRUCTURAL STEEL BY SOLAR PANEL MANUFACTURER.

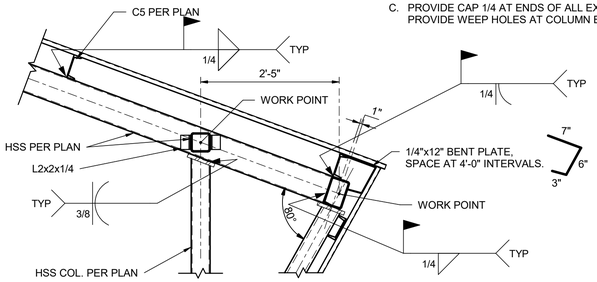
C. PROVIDE CAP 1/4 AT ENDS OF ALL EXPOSED HSS ENDS. PROVIDE WEEP HOLES AT COLUMN BASES AS NEEDED.



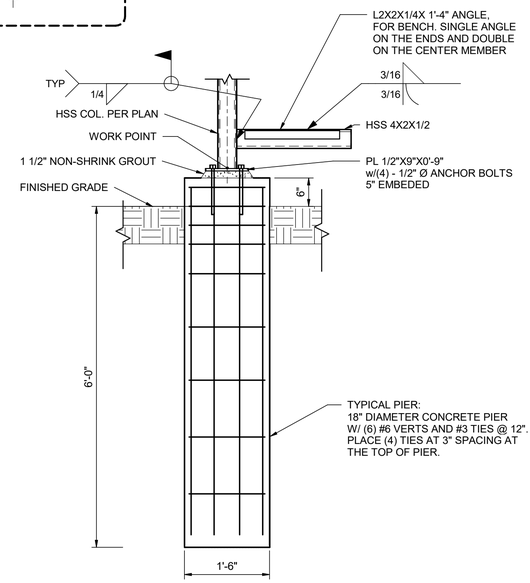
A 121S501 SECTION
SCALE (A)



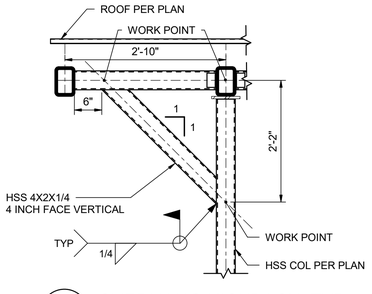
1 121S501 RIDGE DETAIL
SCALE (B)



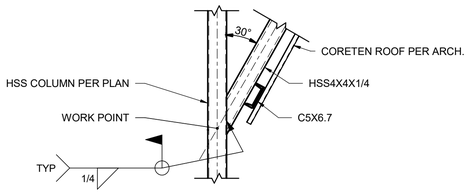
2 121S501 EAVE DETAIL
SCALE (B)



3 121S501 CONCRETE PIER DETAIL
SCALE (B)



4 121S501 OVERHANG KICKBACK, TYP
SCALE (A)



5 121S501 BRACE CONNECTION DETAIL
SCALE (A)

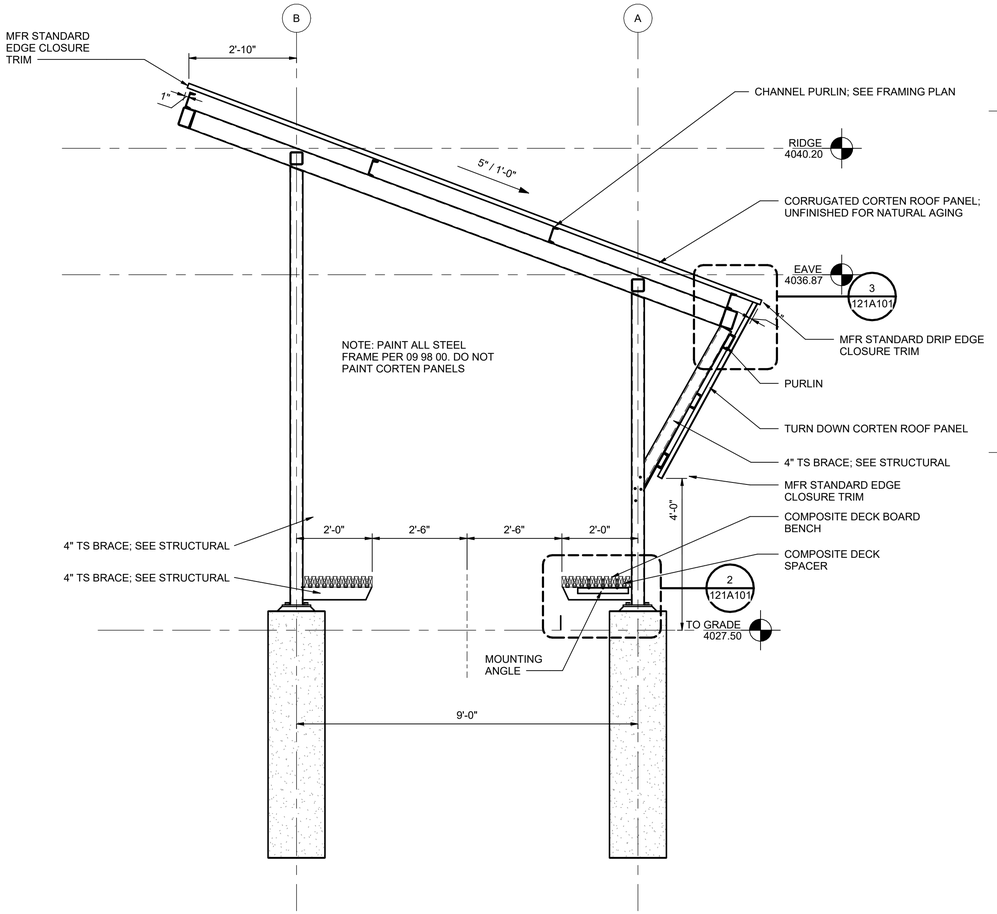
SCALE (B) 1 0 1 2 3 — SCALE OF FEET — 3/4" = 1'-0"

SCALE (A) 2 0 2 4 — SCALE OF FEET — 1/2" = 1'-0"

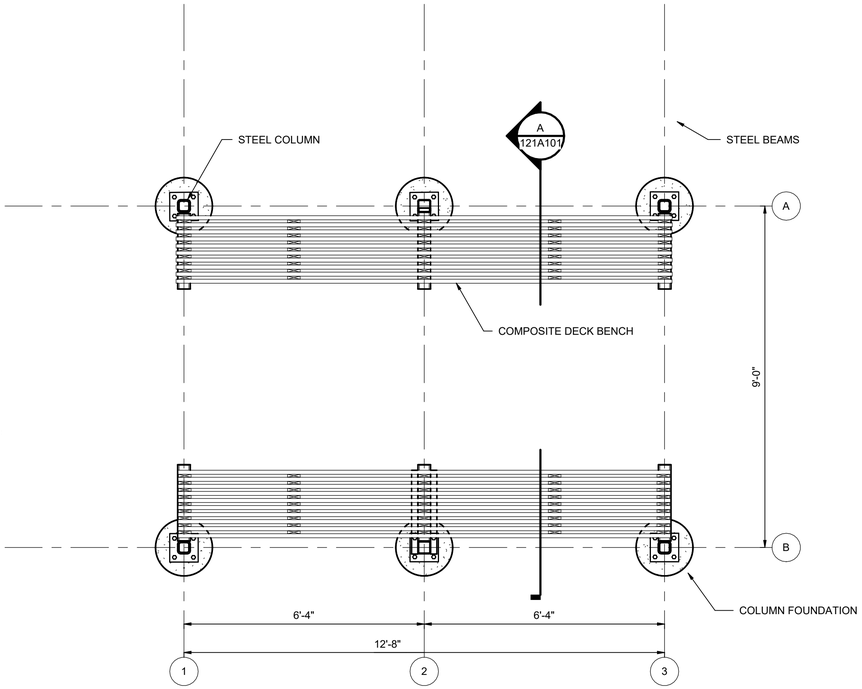
REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 85509 JULIAN CORONADO Date signed effective ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. PEREZ CADD J. PEREZ TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 121 S501	TITLE OF SHEET PHOTOVOLTAIC SYSTEM SOLAR SHADE CANOPY SECTION AND DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 6 OF 11

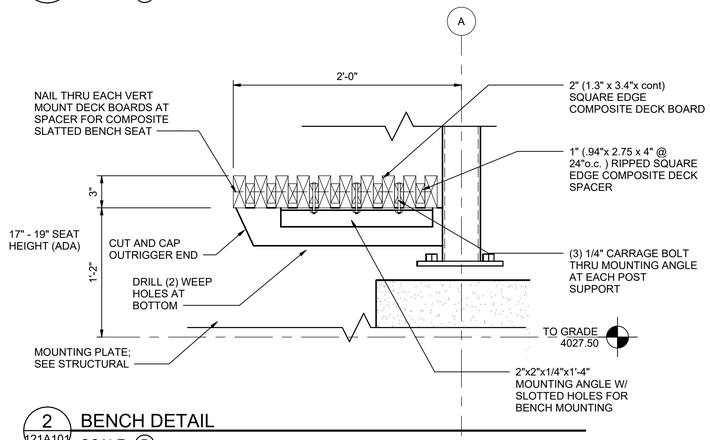
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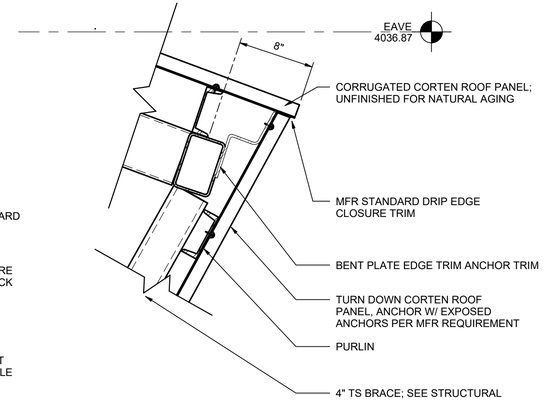
A A101/2 SHADE STRUCTURE SECTION
121A101
SCALE (A)



1 SHADE STRUCTURE PLAN
121A101
SCALE (A)



2 BENCH DETAIL
121A101
SCALE (B)



3 EAVE DETAIL
121A101
SCALE (B)

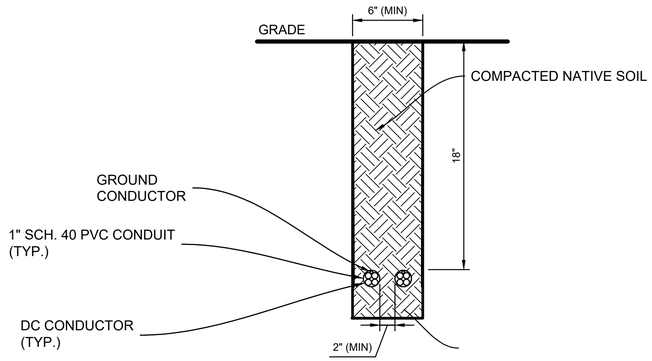
SCALE (B) 12 6 0 12 1-1/2" = 1'-0"
SCALE OF FEET

SCALE (A) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85788 RICHARD MCKINLEY DATE SIGNED 2026.04.21 ARIZONA U.S.A.

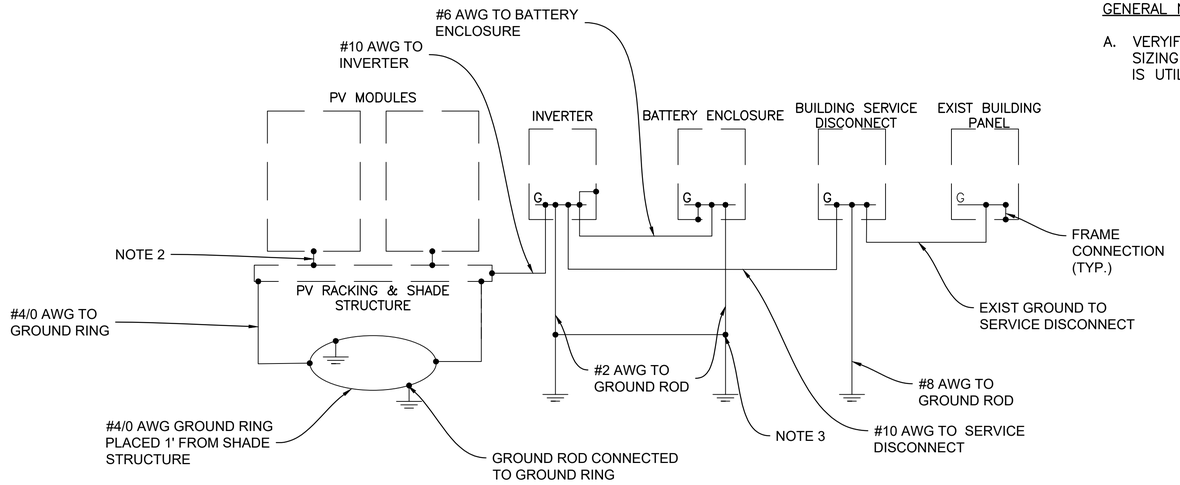
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: R. McKINLEY DRAWN: D. SPERRY TECH. REVIEW: S. HEANEY DATE: MAY 2026	SUB SHEET NO. 121 A101	TITLE OF SHEET PHOTOVOLTAIC SYSTEM SOLAR SHADE CANOPY ARCHITECTURAL PLAN AND SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 7 OF 11

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1 121E501 TRENCH SECTION - PV DC CIRCUIT

SCALE NTS



2 121E501 GROUNDING DIAGRAM

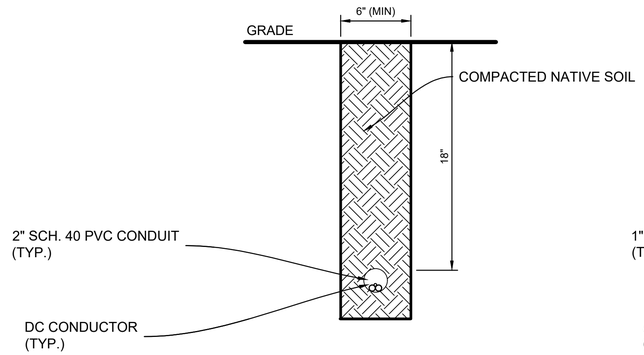
SCALE NTS

GENERAL NOTES:

A. VERYIFY CONDUIT AND CONDUCTOR SIZING IF ALTERNATE EQUIPMENT IS UTILIZED.

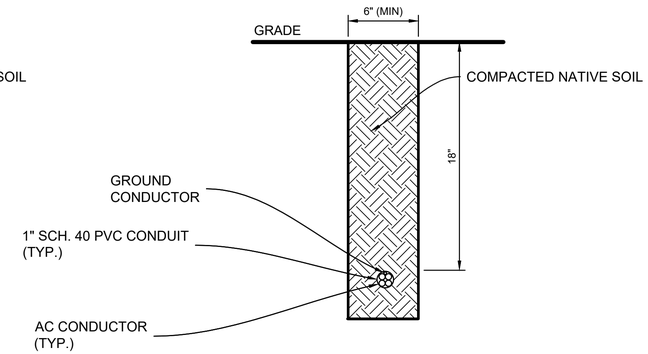
NOTES:

1. SHADE STRUCTURE SHALL BE CONTINUOUSLY BONDED BY SEPARATE GROUND BONDS OR MECHANICAL CONNECTIONS UL-LISTED FOR THE PURPOSE. BOND ALL STRUCTURAL MEMBERS, SUPPORT PILES, AND CONCRETE REBAR.
2. BOND PV MODULE FRAMES TO MODULE RACKING AND STRUCTURE WITH "WEEB" CLIPS OR EQUIVALENT.
3. BURIED GROUND CONNECTIONS SHALL BE EXOTHERMIC WELDS OR OTHER IRREVERSIBLE CONNECTION.
4. BOND EQUIPMENT GROUND TO EQUIPMENT PAD REBAR WHERE APPLICABLE.



3 121E501 TRENCH SECTION - BESS DC CIRCUIT

SCALE NTS



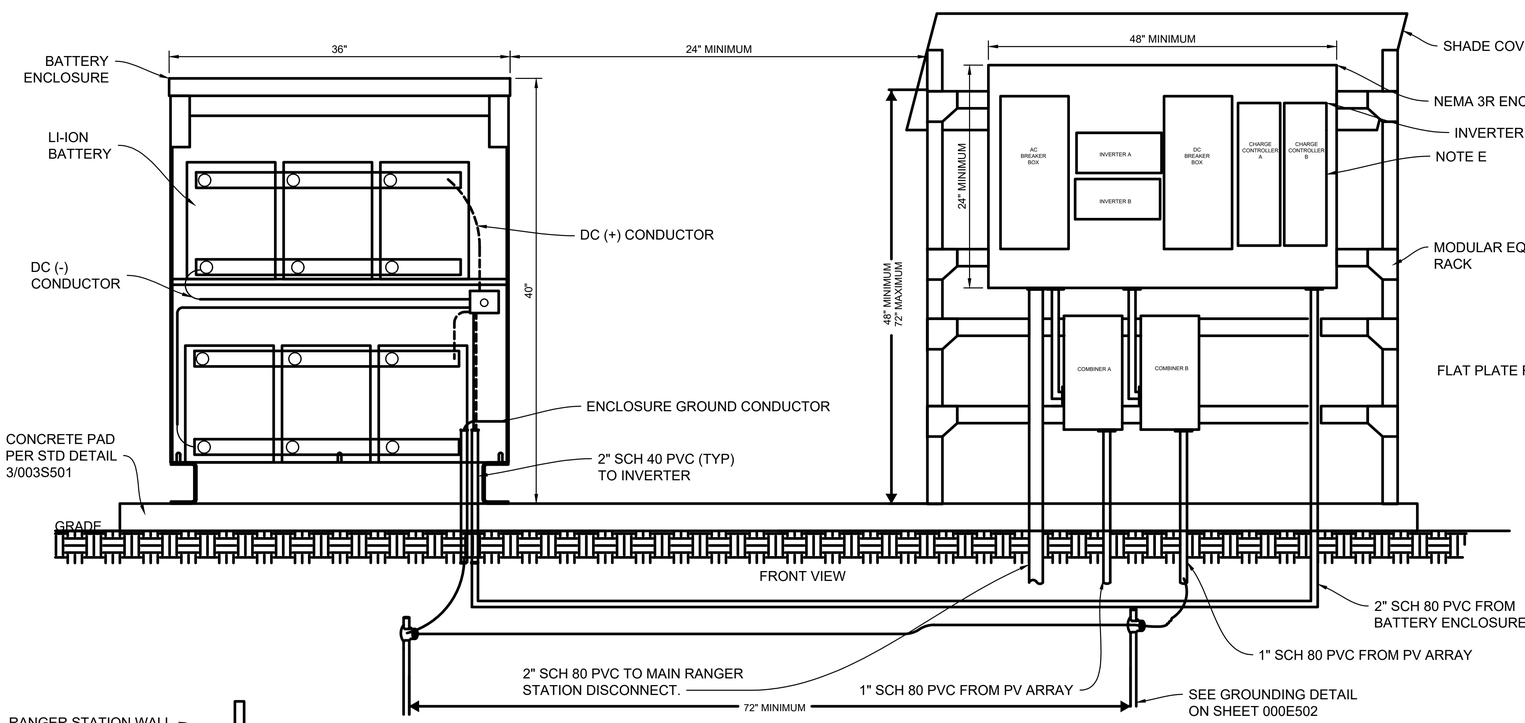
4 121E501 TRENCH SECTION - AC CIRCUIT

SCALE NTS

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

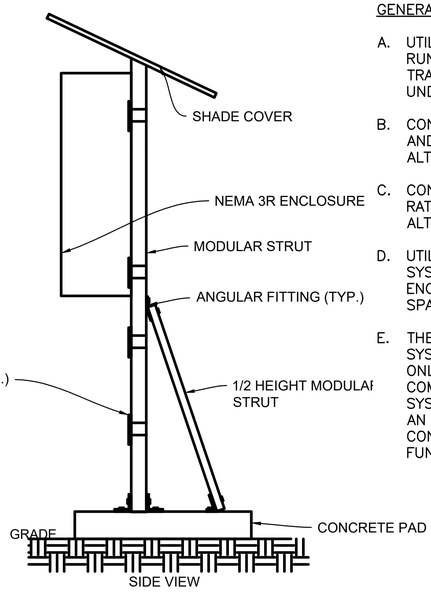
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. KUCHLE CADD D. VANG TECH. REVIEW: A. KANER DATE: MAY 2026	SUB SHEET NO. 121 E501	TITLE OF SHEET PHOTOVOLTAIC SYSTEM GROUNDING AND TRENCH DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 8 OF 11

4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\121-E501.dwg



1 121E502 **BATTERY AND INVERTER ENCLOSURE CONNECTION DETAILS**

SCALE NTS



GENERAL NOTES:

A. UTILIZE SCH 80 PVC FOR CONDUIT RUNS EXPOSED TO SUNLIGHT. TRANSITION TO SCH 40 FOR UNDERGROUND.

B. CONTRACTOR TO VERIFY BATTERY AND INVERTER ENCLOSURE SIZE IF ALTERNATE EQUIPMENT IS UTILIZED.

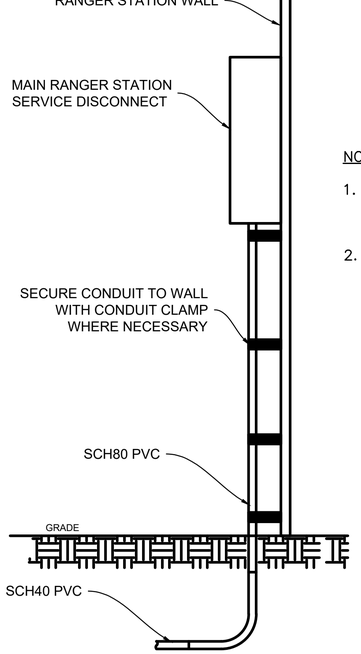
C. CONTRACTOR TO VERIFY EQUIPMENT RATING AND CONNECTIONS IF ALTERNATE EQUIPMENT IS UTILIZED

D. UTILIZE PRE-WIRED INVERTER SYSTEM AND INSTALL IN NEMA 3R ENCLOSURE. VERIFY EQUIPMENT SPACING PRIOR TO INSTALLATION.

E. THE INVERTER AND BATTERY SYSTEM IS SHOWN FOR REFERENCE ONLY TO INDICATE THE REQUIRED COMPONENTS OF THE SYSTEM. SYSTEM SHALL BE DESIGNED BY AN INTEGRATOR AND PROVIDED BY CONTRACTOR AS A FULLY FUNCTIONING SYSTEM.

NOTES:

1. COMBINED EQUIPMENT LOADS PER 36" SPAN SHALL NOT EXCEED 500 LBS.
2. MODULAR STRUT WIDTH: 1 5/8"
3. RACK ASSEMBLY MATERIAL TO MATCH NPS FEATURE REQUIREMENTS. CONFIRM MATERIAL WITH CONTRACT OFFICER.
4. PROTECT SURFACES WITH DISSIMILAR MATERIALS.
5. REPAIR CUT ENDS AND DAMAGED SURFACES.
6. PROVIDE CONDUIT STUB-UP THROUGH CONCRETE PAD INTO BATTERY ENCLOSURE.

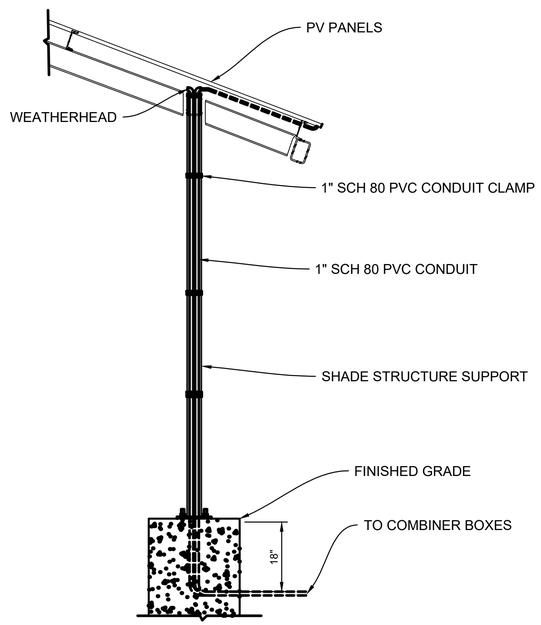


NOTES:

1. CONTRACTOR TO CONFIRM LOCATION AND HEIGHT OF EXIST DISCONNECT PANEL.
2. INSTALL CIRCUIT BREAKER ENCLOSURE ADJACENT TO EXIST SERVICE DISCONNECT ENCLOSURE. DISCONNECT EXIST SECONDARY CONDUCTOR FROM DISCONNECT PANEL AND ROUTE TO NEW CIRCUIT BREAKER ENCLOSURE.

2 121E502 **SERVICE DISCONNECT CONNECTION**

SCALE NTS



NOTES:

1. CONTRACTOR RESPONSIBLE FOR PV PANEL MOUNTING AND ASSOCIATED DESIGN. ARRAY OUTPUT CONDUCTORS TO TRANSITION INTO CONDUIT AT CORNER OF SHADE STRUCTURE.
2. DETAIL IS DIAGRAMMATIC TO INDICATE CONDUIT ROUTING. CONDUIT TO BE SURFACE MOUNTED. SEE SHEET 121S501 FOR STRUCTURAL DETAILS.

3 121E502 **PV WIRING AND CONDUIT ROUTING**

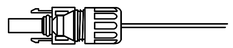
SCALE NTS

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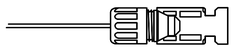
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. KUCHLE DRAWN: S. BILD TECH. REVIEW: A. KANER DATE: MAY 2026	SUB SHEET NO. 121 E502	TITLE OF SHEET PHOTOVOLTAIC SYSTEM SOLAR / BATTERY DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 9 OF 11

4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\121-E502.dwg

FEMALE CONNECTOR



MALE CONNECTOR

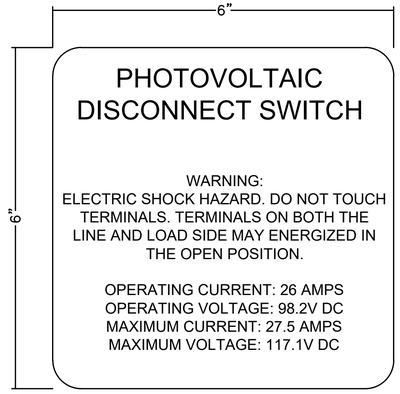


CONNECTORS SHALL BE:

1. OF THE SAME MANUFACTURER
2. LOCKING TYPE, 1000VDC
3. COMPATIBLE WITH MANUFACTURER'S SUPPLIED CONNECTORS
4. CRIMPED ONTO THE SOURCE CIRCUIT WIRE USING PROCEDURE
5. TOOLS RECOMMENDED BY MANUFACTURER
6. SECURED SO NOT SUBJECTED TO RAIN DIRECTLY

1 / 121E503 — PV MODULE CONNECTOR DETAIL

SCALE NTS

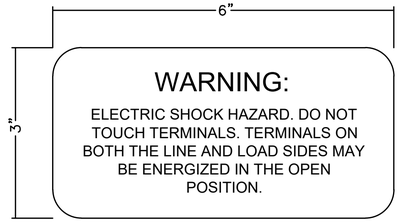


2 / 121E503 — PV DISCONNECT SWITCH WARNING SIGN

SCALE NTS

DETAILS NOTES:

1. PROVIDE WARNING SIGN FOR EACH PV COMBINER/DISCONNECT.

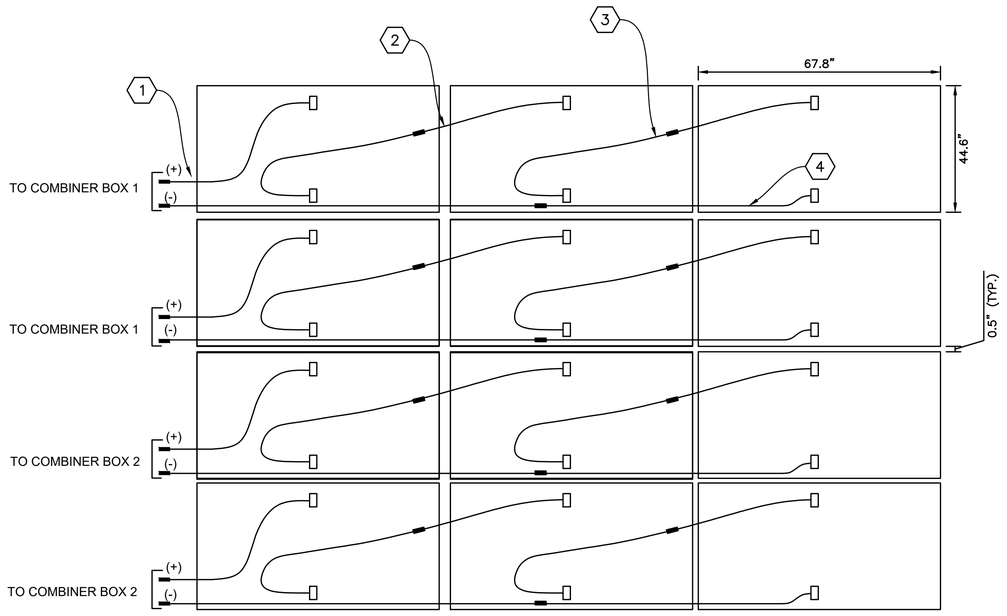


3 / 121E503 — PV DISCONNECT SWITCH WARNING SIGN

SCALE NTS

DETAILS NOTES:

1. PROVIDE WARNING LABEL FOR BATTERY ENCLOSURE AND INVERTER UNIT.



4 / 121E503 — TYPICAL PV WIRE MANAGEMENT

SCALE NTS

DETAILS NOTES:

1. DO NOT STRAIN MANUFACTURER INSTALLED MODULE LEADS.
2. SECURE WIRES TO MODULE FRAME WITH STAINLESS STEEL METAL CLIP. CLIPS SHALL BE SIZED APPROPRIATELY TO SECURELY FASTEN THE CABLE SIZE/QUANTITY AS SPECIFIED IN THE DESIGN.
3. ENSURE CONNECTORS ARE UNDER THE MODULES, PROTECTED FROM ELEMENTS, NOT IN THE GAP BETWEEN MODULES.

GENERAL NOTES:

A. CONTRACTOR TO PROVIDE WARNING LABELS OR PLACARDS AS REQUIRED BY THE NEC AND AHJ.

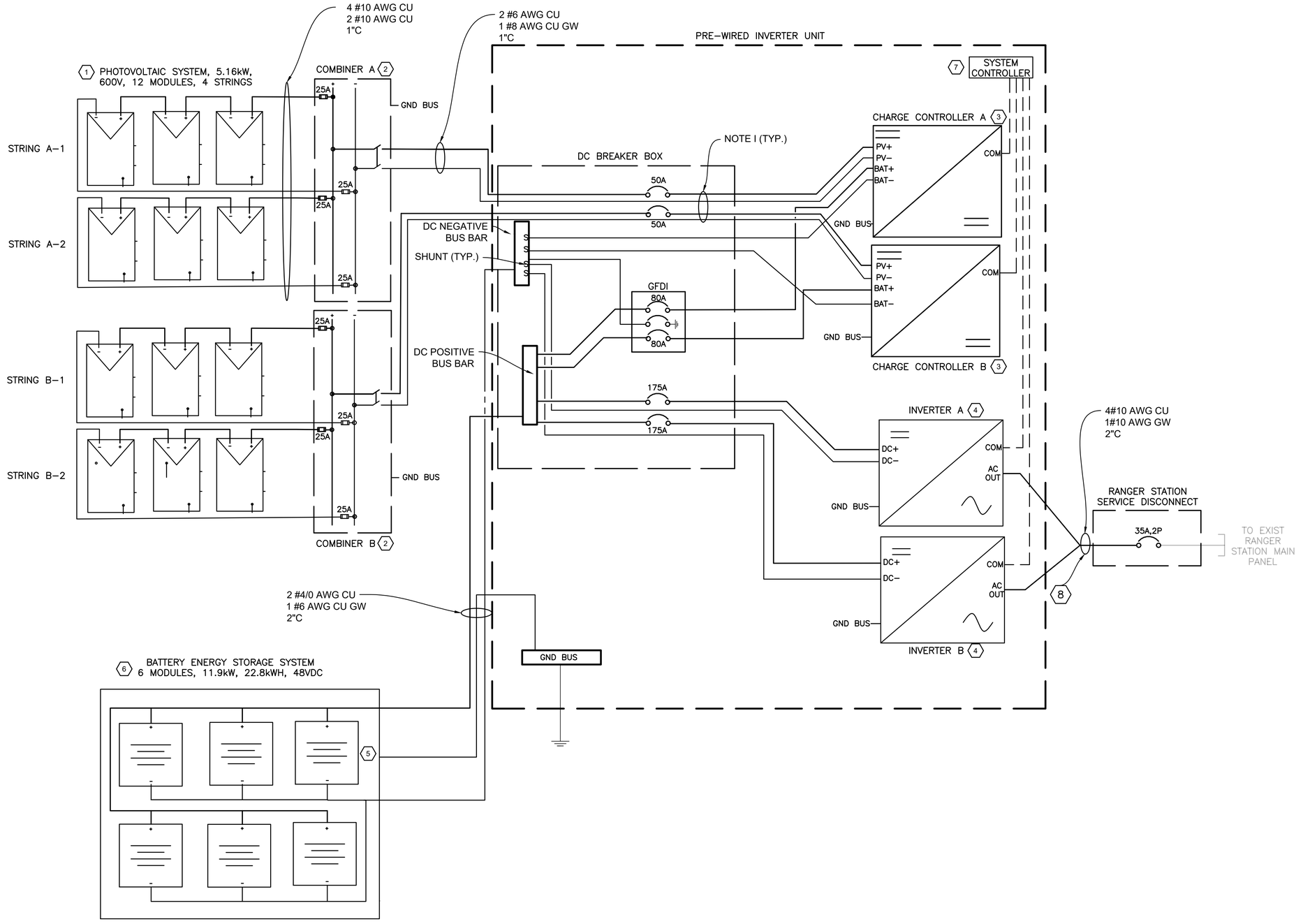
KEYNOTES: (X)

1. KEEP (+) AND (–) TOGETHER ON SAME STRING WHERE POSSIBLE.
2. MODULE LEADS SHALL BE CONNECTED TO FRAME WITH CLIPS (TYP).
3. JUMPERS SHALL BE FURNISHED AS REQUIRED.
4. ROUTE STRING HOMERUN ALONG MODULE AND SECURE EVERY 48" ALONG MODULE FRAME OR RACKING WITH METAL CABLE TIES OR EQUIVALENT FASTENER.

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75014 — TOU VANG — STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. KUCHLE CADD: S. BILD TECH. REVIEW: A. KANER DATE: MAY 2026	SUB SHEET NO. 121 E503	TITLE OF SHEET PHOTOVOLTAIC SYSTEM PV CONNECTION DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 10 OF 11

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GENERAL NOTES:

A. PROVIDE SYSTEM COMPONENTS TO MEET THE PERFORMANCE CHARACTERISTICS IN THE PROJECT DRAWINGS AND SPECIFICATIONS.

B. EQUIPMENT REPRESENTED IN VOLUME 12 DESIGN DOCUMENTS AS BASIS OF DESIGN MAY BE SUBSTITUTED WITH EQUIVALENT PER SPECIFICATIONS. CONTRACTOR IS RESPONSIBLE FOR ANY REQUIRED REDESIGN.

C. ALL EQUIPMENT SHALL WORK TOGETHER TO FORM A FULLY FUNCTIONING SYSTEM WITHOUT ADJUSTMENT, DERATES, OR FIELD MODIFICATION. PV STRING, FUSE, AND WIRE SIZING ARE BASED ON SELECTED COMPONENTS. VERIFY SIZING IF ALTERNATE EQUIPMENT IS UTILIZED.

D. CONDUCTOR TYPES SHALL BE XHHW-2. UTILZIE PV WIRE FOR DC CIRCUITS. CONDUIT TO BE SCH 80 RIGID PVC FOR EXPOSED RUNS AND SCH 40 PVC FOR BURIED RUNS.

E. SEE SHEET 121E501 FOR GROUNDING CONNECTION DETAILS.

F. SEE SHEET 121E502 FOR BATTERY CONNECTIONS DETAILS.

G. SEE SHEET 121E503 FOR PV CONNECTION DETAILS.

H. CONNECT DC NEGATIVE BUS TO INPUT OF GROUND FAULT DETECTION INTERRUPTER (GFDI) 0.5 AMP BREAKER. CONNECT 0.5 AMP BREAKER OUTPUT TO GROUND BUS.

I. INVERTER/CHARGER SYSTEM SHALL HAVE PRE-WIRED INTERNAL CONNECTIONS. INDICATED WIRING IS DIAGRAMMATIC AND MAY VARY FROM INSTALLED EQUIPMENT. VERIFY WIRING PRIOR TO INSTALLATION.

J. THE BATTERY AND INVERTER/CHARGER SYSTEM IS SHOWN FOR REFERENCE ONLY TO INDICATE THE REQUIRED COMPONENTS AND FUNCTIONALITY OF THE SYSTEM. THE SYSTEM SHALL BE DESIGNED BY AN INTEGRATOR AND PROVIDED BY CONTRACTOR AS A FULLY FUNCTIONING SYSTEM WITH ALL REQUIRED ACCESSORIES AND APPURTENANCES.

KEY NOTES: (X)

1. PV MODULES 430 WATT MODULE TO BE USED OR EQUIVALENT PER CHARACTERISTICS DEFINED IN SPECIFICATION 26 31 00. MODULES INCLUDE A JUNCTION BOX WITH POSITIVE AND NEGATIVE LEADS AND CONNECTORS. FIELD INSTALLED CONNECTORS MUST EXACTLY MATCH MODULE CONNECTORS.
2. COMBINER BOX 600 VDC, 100 AMP COMBINER BOX WITH (2) FUSED INPUTS AND 2-POLE DISCONNECT, RATED FOR 100% CONTINUOUS DUTY. COMBINER BOX OUTPUTS SHALL INCLUDE MOISTURE RELIEF VALVE, DC SURGE PROTECTION AND PROVISIONS FOR PADLOCK. COMBINER BOX SHALL BE UL 1741 LISTED.
3. CHARGE CONTROLLER 48 VDC, 80 AMP CHARGE CONTROLLER TO BE USED PER CHARACTERISTICS DEFINED IN SPECIFICATION 26 33 00.
4. INVERTER/CHARGER 3,000 VA, 120 VAC, 48 VDC INVERTER/CHARGER TO BE USED PER CHARACTERISTICS DEFINED IN SPECIFICATION 26 33 00.
5. BATTERY MODULE, 1.9KW PEAK, 3.8 KWH, 48 VDC LI-ION BATTERY MODULE TO BE USED PER CHARACTERISTICS DEFINED IN SPECIFICATION 26 33 00.
6. BATTERY ENCLOSURE PAD-MOUNTED NEMA 3R ENCLOSURE TO HOUSE BATTERY MODULES. ENCLOSURE SHALL INCLUDE PASSIVE ENVIRONMENTAL CONDITIONS TO SUPPORT THE BATTERY MFG. REQUIREMENTS. AUXILIARY POWER SHALL NOT BE REQUIRED FOR ENCLOSURE CONDITIONING.
7. SYSTEM CONTROLLER SHALL BE COMPATIBLE WITH CHARGE CONTROLLER, INVERTER/CHARGER, AND BATTERY SYSTEM PER CHARACTERISTICS DEFINED IN SPECIFICATION 26 33 00.
8. INVERTERS TO OPERATE IN SPLIT-PHASE CONFIGURATION TO OUTPUT 240V 2P TO SERVICE DISCONNECT.

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75014
TOU VANG
STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. KUCHLE CADD R.D. CLARK TECH. REVIEW: A. KANER DATE: MAY 2026	SUB SHEET NO. 121 E601	TITLE OF SHEET PHOTOVOLTAIC SYSTEM DC/AC SCHEMATIC DIAGRAM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 11 OF 11

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